HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** microservices, implementing REST APIs for cross-service data exchange and handling approximately **1,500 requests per minute**.
- Designed and implemented **Hibernate** entities and JPA mappings to manage relational data persistence for core enterprise applications, ensuring data integrity and efficient querying across multiple schemas.
- Engineered **JUnit and Mockito** test suites for Java backend services, increasing code coverage by **18%** and reducing post-deployment defects by identifying critical logic flaws prior to release.
- Contributed to the development and manufacture of life-changing medical technologies by applying engineering principles to solve technical challenges.
- Collaborated with cross-functional teams across Engineering and Quality departments to support projects that improve product quality and patient safety.
- Applied engineering principles to problem-solve and debug issues within a regulated manufacturing environment, ensuring operational continuity.
- Containerized backend Java services using **Docker** and deployed them to **AWS EC2** instances, configuring basic resource limits and network access for predictable performance.

SKILLS

Languages – Java, Python, SQL, R

Backend Frameworks – Spring Boot, Hibernate, JPA, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, R/RStudio

Methodologies – Agile, Scrum

Tools – Jira, Confluence, Splunk, Dynatrace

PROJECTS

Enterprise Java Application Modernization (HCL Technologies)

- Modernized a legacy Java application by re-architecting core modules into **Spring Boot microservices**, improving system modularity and maintainability for an enterprise client.
- Implemented robust data persistence layers using **Hibernate and PostgreSQL**, managing complex object-relational mappings and optimizing query performance for high-volume transaction data.
- Developed comprehensive unit and integration tests with **JUnit and Mockito**, achieving over **85% code coverage** for critical business logic and reducing regression testing cycles by **30%**.

Medical Device Manufacturing Process Analytics (MSc Project)

- Developed analytical models in **Python** using **pandas and scikit-learn** to analyze manufacturing process data, identifying key factors affecting product yield.
- Built interactive **Power BI dashboards** to visualize process variations and key performance indicators for manufacturing stakeholders, enabling data-informed decision-making.
- Applied statistical analysis techniques, including **regression modeling**, to quantify the impact of process parameters on device quality, achieving a model R-squared of **0.78**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)